

Pasture, Rangeland, and Forage Insurance: Benefits and Strategies for Iowa's Pastures

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Pasture, Rangeland, and Forage Insurance (PRF)

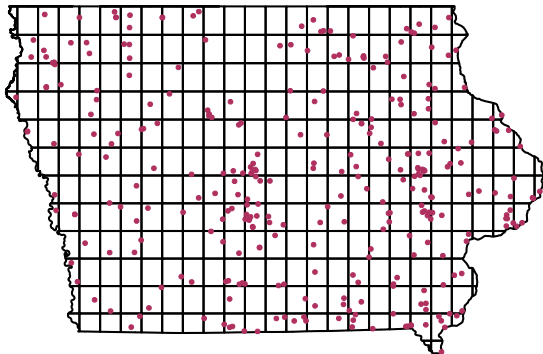
- ▶ Rainfall-index based insurance product offered by the USDA RMA
 - ▷ Available in Iowa since 2016
- ▶ Covers increased costs to supplement reduced forage yields in drier than usual conditions
- ▶ Shares some design with other USDA RMA buy-up coverage insurance products, area risk protection insurance (ARPI)
- ▶ Premiums are subsidized 51-59%

Rainfall Index

- ▶ Daily precipitation from NOAA (1948 - present) averaged over a grid
 - ▷ weighted across nearest 4 stations
- ▶ Index uses total precipitation for 11 bimonthly intervals (Jan-Feb, Feb-Mar,..., Nov-Dec)
- ▶ For a given year, grid, and bimonthly intervals, rainfall index is

$$\frac{\text{Total Precip}}{\text{Mean Precip (1948 to present - 2 years)}}$$

Grids and Weather Stations



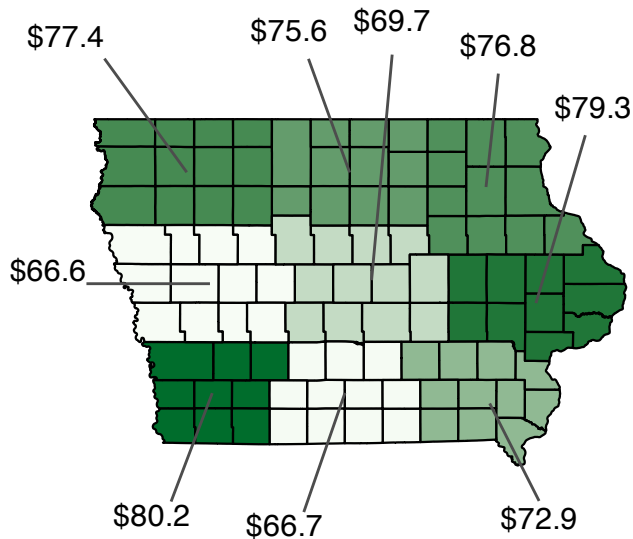
*each grid is $\approx$ 17 by 12 miles

PRF: Insured Value

- ▶ Insured value is determined by your acreage, intended use, county base value, and coverage level
- ▶ Not required to insure all acreage
- ▶ Intended use can be Hay or Grazing
 - ▷ Haying uses separated by irrigation practice, organic, transitional
- ▶ County base value
 - ▷ average “price of alternative feeding” per acre for a county
 - ▷ Determined by FCIC, revised every 3 to 5 years
 - ▷ Can adjust using productivity if your pasture is more or less productive than the county average

2026 County Base Values - Grazing

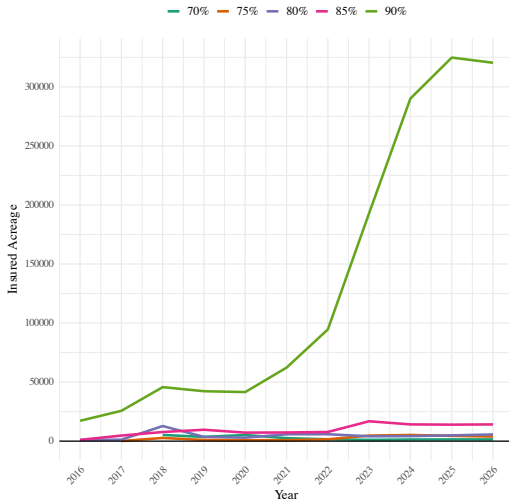
- County base values range from \$66.6 to \$80.2 per acre for grazing



PRF: Coverage Level

- ▶ 70%, 75%, 80%, 85%, 90%
- ▶ “trigger” point for how low the rainfall index must be before an indemnity is paid
- ▶ For a 90% coverage level with \$10,000 insured, if the rainfall index is 85% then you would receive an indemnity payment of

$$(0.90 - 0.85) \times \$10,000 = \$500$$



PRF: Interval Selection

- ▶ You choose a set of intervals and assign a percent of value (%) to each
 - ▷ 60% May-Jun + 40% July-Aug
 - ▷ % of value is how much of your total insured value should be covered in that bimonthly interval
- ▶ Maximum % of value is 60%, minimum is 10%, must total 100%
- ▶ Cannot overlap months

State

Iowa

County

Decatur

Grid ID

24745

OR

Protection Information

Intended Use

Grazing

Irrigation Practice

Please Select

Organic Practice

Please Select

Coverage Level

85%

Productivity Factor

100%

Insurable Interest

100%

Insured Acres

300

Sample Year

2025

Policy Information

County Base Value

\$66.70

Dollar Amount of Protection

\$56.70

Total Insured Acres

300

Protection Table

Index Interval	Percent of Value (%)	Policy Protection Per Unit
Jan-Feb	N/A	\$0
Feb-Mar	N/A	\$0
Mar-Apr	N/A	\$0
Apr-May	N/A	\$0
May-Jun	60	\$10,206
Jun-Jul	N/A	\$0
Jul-Aug	40	\$6,804
Aug-Sep	N/A	\$0
Sep-Oct	N/A	\$0
Oct-Nov	N/A	\$0
Nov-Dec	N/A	\$0
Per Acre	N/A	N/A
Total	300	\$17,010

PRF: Grid Selection

- ▶ If your acreage covers multiple grids you have two options:
 - ▷ Consolidate under one grid
 - ▷ Separate acreages, separate policies
- ▶ Grid selection in these instances can be used to hedge risk, pick a lower premium rate, etc.
- ▶ Cannot consolidate across counties

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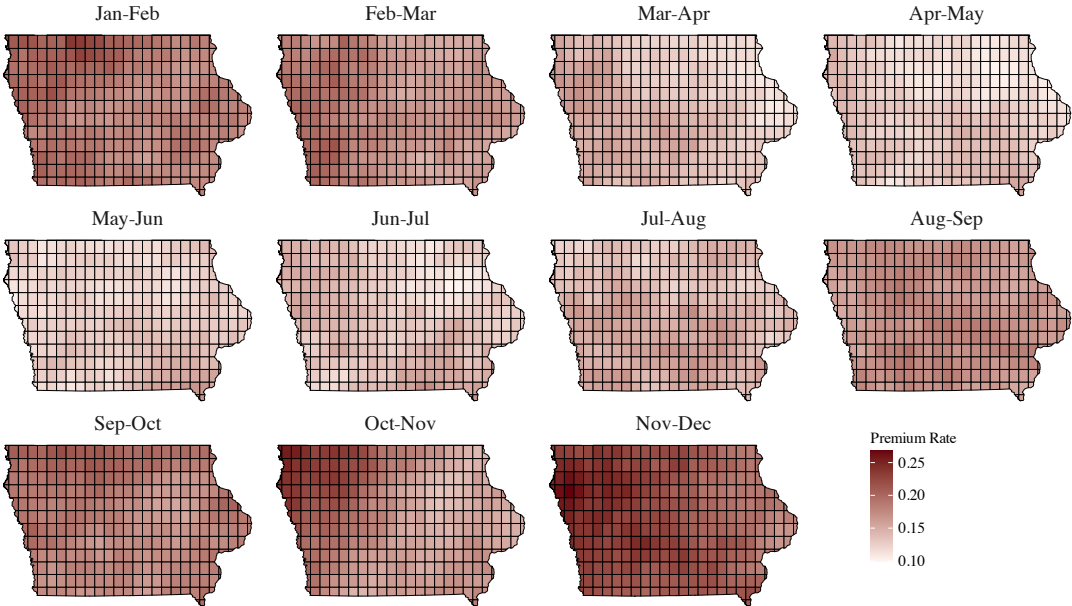
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PRF: Premiums

- ▶ Premiums depend on “where”, “when”, and “how much” you insure.
- ▶ USDA RMA sets premium rates every year for each grid, bimonthly interval, intended use, and coverage level
 - ▷ They use actuarially fair rates (premium $\approx$ average indemnity)
 - ▷ But add approximately 14% for “catastrophic loading”
- ▶ Roughly based on how risky it is to insure.
 - ▷ If rainfall is more variable year to year in grid and/or bimonthly interval then its premium rate will be higher.
 - ▷ Higher coverage levels have higher premium rates because the trigger point is higher.

2026 Premium Rates - Grazing



PRF: Pros and Cons

► Pros

- ▷ **Extremely flexible.** You can configure a contract to fit your operation and goals to a considerable degree
- ▷ **Quicker indemnity payments.** You'll receive any indemnities approx. 2 - 3 months after the insured interval.
- ▷ **No upfront costs.** Indemnities are first credited to total premium. If you need to pay, deadline isn't until Sep. 1 of the subsequent crop year.
- ▷ **Subsidized.** On average, from 2016 to 2025 contracts in Iowa returned:

\$1.56 in indemnity per \$1 paid in premium.

* but not guaranteed

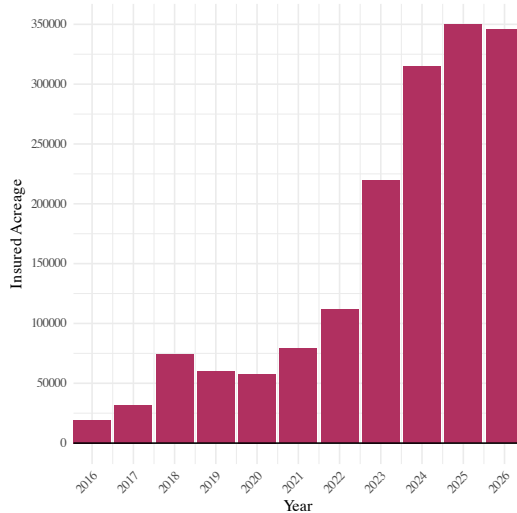
PRF: Pros and Cons

► Cons

- ▷ **Basis Risk.** The rainfall index can poorly correlate with forage outcomes.
 - ◇ Falsely assumes a linear relationship.
 - ◇ Rainfall index may fail to trigger an indemnity in some bad forage years.
- ▷ **Complexity.** Many possible combinations of intervals. Optimizing choices can be challenging and time consuming.
- ▷ **Single Peril.** Only protects from a lack of rainfall, not other drought or flood conditions.

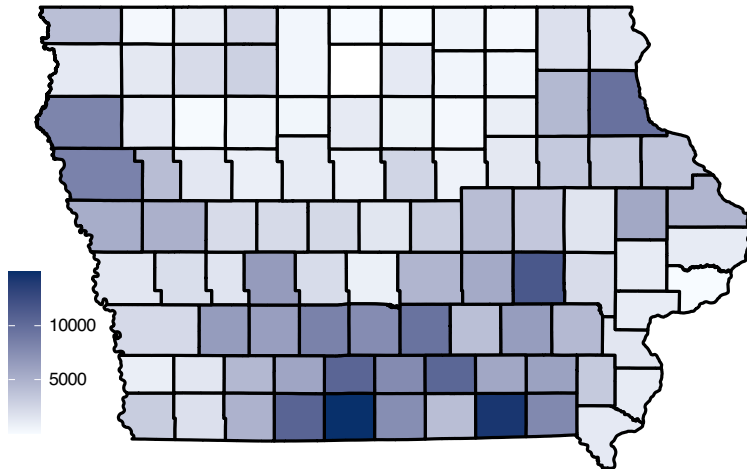
How is PRF being used in Iowa?

► Insured Acreage by Year



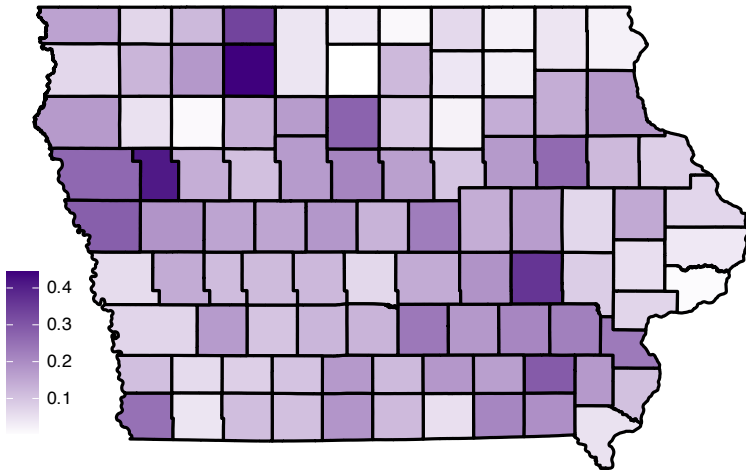
How is PRF being used in Iowa?

► Insured Acreage by County - 2025



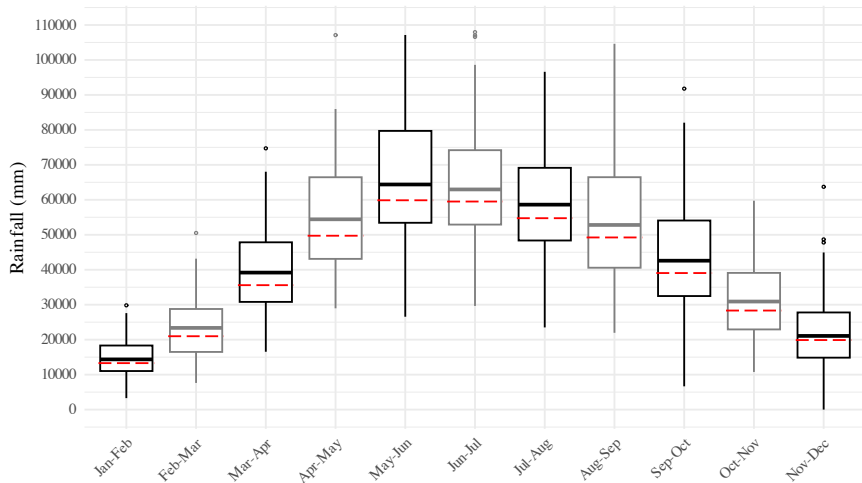
How is PRF being used in Iowa?

► Share of Eligible Acres Insured by County



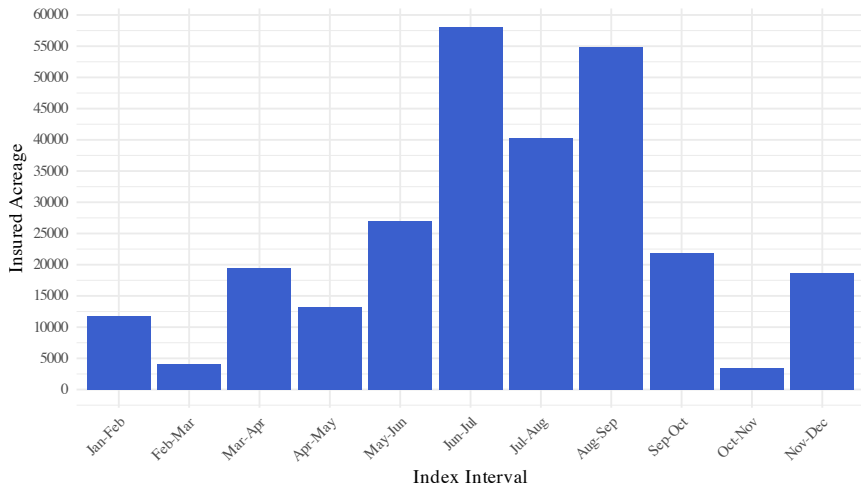
How is PRF being used in Iowa?

► Total Rainfall Distribution Across Index Intervals - **Iowa**



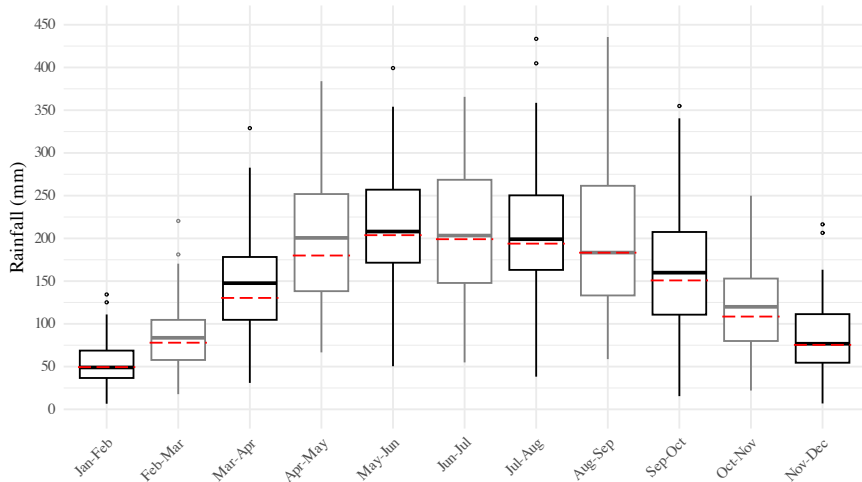
How is PRF being used in Iowa?

► Insured Acreage by Index Interval - Iowa



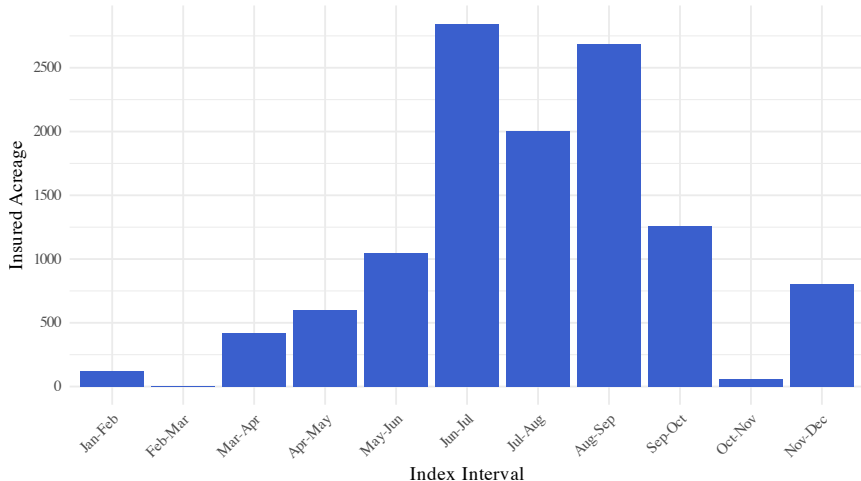
How is PRF being used in Iowa?

► Rainfall Distribution Across Intervals - Decatur County grid 24745



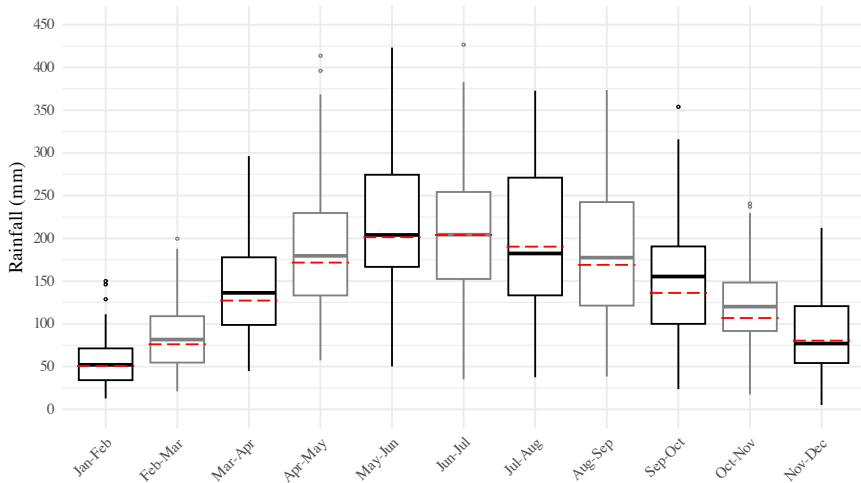
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► Insured Acreage by Index Interval - Decatur County



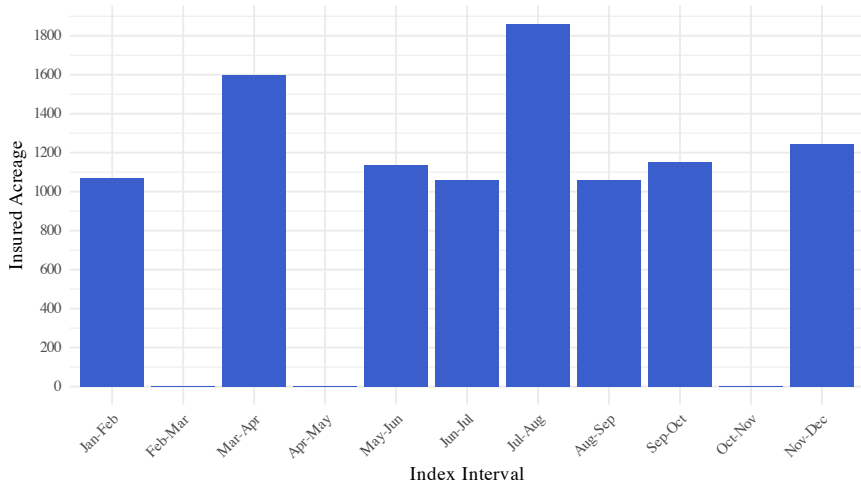
How is PRF being used in Iowa?

► Rainfall Distribution Across Intervals - **Iowa County** grid 25952



How is PRF being used in Iowa?

► Insured Acreage by Index Interval - Iowa County



PRF Interval Selection Strategies

- ▶ Anecdotally, we see considerable diversity in intervals insured. Why?
- ▶ Theory: Farmers use PRF in different ways to satisfy their unique goals and risk preferences
- ▶ At least two distinct strategies:
 - ▷ **Risk management.** Select intervals with indexes that correlate well with forage outcomes.
 - ▷ **Profit maximization.** Select intervals that generate the highest return, even if it means selecting “riskier” intervals.

Simulation

- ▶ Simulate insurance outcomes for a large set of interval combinations
 - ▷ % of value in increments of 5%
 - ▷ $\approx 10,000$ contracts
- ▶ Loss Ratio = $\frac{\text{Avg. Indemnity}}{\text{Premium (subsidized)}}$
- ▶ Compare with how frequently the contract paid out in years with avg. pasture condition $\geq 20\%$ poor or very poor
 - ▷ USDA NASS 1995-2025, entire state

County	Decatur
Grid	24745
Variable	Value
Stocking Rate	3
Cow/Calf Pairs	20
Acres	60
Productivity Factor	1
Coverage Level	90%

Risk Management Strategy

Top 5 Contracts - Risk Management

Contract	Premium	Avg. Indemnity	Loss Ratio	Freq. Pay in Bad Cond.
Mar-Apr + Jun-Jul + Sep-Oct 0.1 + 0.3 + 0.6	\$288.58	\$602.31	2.09	1
Mar-Apr + Jun-Jul + Sep-Oct 0.1 + 0.4 + 0.5	\$281.85	\$582.39	2.07	1
Mar-Apr + Jun-Jul + Sep-Oct 0.1 + 0.5 + 0.4	\$275.13	\$562.47	2.04	1
Mar-Apr + Jun-Jul + Sep-Oct 0.2 + 0.2 + 0.6	\$290.43	\$591.06	2.04	1
Mar-Apr + Jun-Jul + Sep-Oct 0.1 + 0.6 + 0.3	\$268.40	\$542.56	2.02	1
Mar-Apr + Jun-Jul + Sep-Oct 0.2 + 0.3 + 0.5	\$283.70	\$571.14	2.01	1

Profit Maximization Strategy

Top 5 Contracts - Profit Maximization

Contract	Premium	Avg. Indemnity	Loss Ratio	Freq. Pay in Bad Cond.
Jun-Jul + Sep-Oct 0.4 + 0.6	\$286.72	\$613.57	2.14	0.8
Jun-Jul + Sep-Oct 0.5 + 0.5	\$280.00	\$593.65	2.12	0.8
Jun-Jul + Sep-Oct 0.6 + 0.4	\$273.27	\$573.73	2.10	0.8
Mar-Apr + Jun-Jul + Sep-Oct 0.1 + 0.3 + 0.6	\$288.58	\$602.31	2.09	1.0
Apr-May + Jun-Jul + Sep-Oct 0.1 + 0.3 + 0.6	\$284.87	\$592.90	2.08	0.9
Mar-Apr + Jun-Jul + Sep-Oct 0.1 + 0.4 + 0.5	\$281.85	\$582.39	2.07	1.0

Which Interval Selection Strategy to Use?

- ▶ Tradeoff between risk exposure and returns.
 - ▷ Tradeoff is less significant in IA compared to other states
- ▶ Premium subsidy can incentivize selecting riskier intervals.
- ▶ What will work best for you depends on your goals, appetite for risk, pasture location, forage system.
- ▶ Many excellent resources are available to support you!

Additional PRF Resources

- ▶ USDA PRF Support Tool
 - ▷ Find your grid(s)
 - ▷ See historical index performance
 - ▷ Estimate premiums and indemnities across different contract choices (e.g. intervals, coverage, use, etc.)
- ▶ Iowa Farm Bureau
- ▶ Your local insurance agent
- ▶ Iowa State Ag Decision Maker

Key Takeaways

- ▶ PRF is an effective, flexible, and subsidized insurance policy.
- ▶ But it's underutilized.
- ▶ Iowan's are using various, well-informed strategies to make the most of it.

Scan for a 1 page summary!

